

GROUP - E

- IX. (a) Explain different methods of compensation used in control systems.
 (b) Discuss the relative merits of proportional, P I, and P.I.D. controllers.

OR

- X. (a) Explain the use of lag, lead and lag-lead compensation networks.
 (b) What are the constraints on the time constants of lag and lead sections of the lag-lead network? Explain.

**B.TECH. DEGREE VI SEMESTER EXAMINATION IN
 MECHANICAL ENGINEERING (CAD/CAM)
 NOVEMBER 2001**

ME 605 CONTROL SYSTEMS*(1995 Admissions)*

Time: 3 Hours

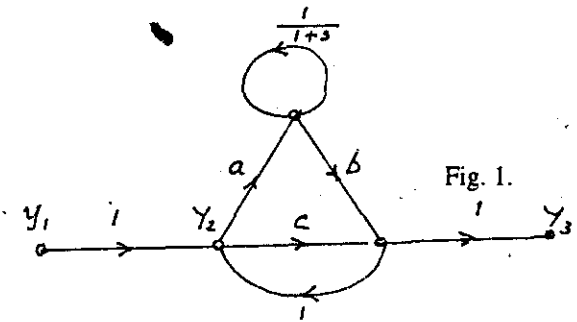
Maximum Marks: 100

(All questions carry **EQUAL** marks)**GROUP - A**

- I. (a) Explain the unit impulse function $\delta(t)$ and derive its Laplace transform.
 (b) How is the impulse response of a system related to its transfer function?
 (c) Assuming necessary parameters find the transfer function of an armature controlled d.c. motor.

OR

- II. (a) A servomechanism is used to control the angular position of a mass θ_o , through a command signal θ_i . The moment of inertia of the moving parts is 200 Kgm^2 , the motor torque $6.88 \times 10^4 \text{ Nm}$ per rad of error, and the damping torque coefficient $5 \times 10^3 \text{ Nm}$ per rad/s, all referred to the load shaft. Form the equation for the dynamic system and obtain the transfer function $\frac{\theta_o(s)}{\theta_i(s)}$.
 (b) For the system represented by signal flow graph in figure 1, determine the transfer functions (i) $\frac{y_3(s)}{y_1(s)}$ and (ii) $\frac{y_2(s)}{y_1(s)}$, using Mason's formula.



GROUP - B

- III. (a) A unity feed back system has forward transfer function
 $G(s) = \frac{12}{s(2s+11)}$. Obtain the response of the system to unit impulse input.
- (b) For the system represented in figure 2, determine the steady state error when the command signal is a constant angular velocity of 1 rev/minute.

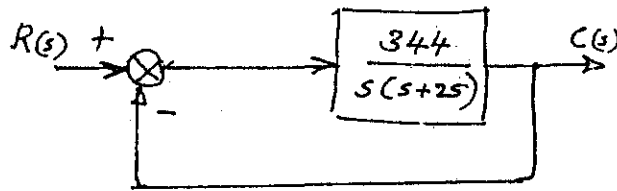


Fig. 2

OR

- IV. (a) Explain steady state error constants K_p , K_v and K_a .
- (b) The open loop transfer function of a servosystem with unity feed back is

$$G(s) = \frac{10}{s(1+0.1s)}$$

Evaluate the dynamic error using the dynamic error coefficients.

GROUP - C

- V. (a) What are the desirable characteristics of a servomotor? Describe the steps adopted to approach these characteristics in the design of an a.c. servomotor.

- V. (b) For the system whose characteristic equation is given below find the range of K for which the system is stable, if such range exists.

$$s^4 + Ks^3 + 2s^2 + (K+1)s + 10 = 0$$

OR

- VI. (a) What is the principle of argument in connection with the plotting of the root-locus for $0 < K < \infty$? How is the principle applied in determining the angle of departure and the angle of arrival?
- (b) The characteristic equation of a feed back system is given by $s^2 + (2+K)s + 2(1+K) = 0$. Sketch the root locus for $0 < K < \infty$. What is the range of K for which the transient response has oscillatory component?

GROUP - D

- VII. (a) State and explain Nyquist's stability criterion.
- (b) By the use of Nyquist's criterion determine whether the closed loop system having the following loop transfer function is stable or not.

$$G(s)H(s) = \frac{1+4s}{s^2(1+s)(1+2s)}$$

If unstable how many roots of the characteristic equation lie on the right half s -plane?

OR

- VIII. (a) Explain the systematic procedure for obtaining the Bode-plot of the loop transfer function of a feed back system. Comment on the inaccuracies of the asymptotic magnitude plot.
- (b) How is the stability of the closed loop system predicted from examination of the Bode-plot?

Contd.....4.

Contd.....3.